Journal of Pediatric Nursing 81 (2025) 8–15

Contents lists available at ScienceDirect

Journal of Pediatric Nursing

journal homepage: www.pediatricnursing.org

# Emotional and psychosocial functioning in youngsters with a congenital heart disease (CHD) in comparison to healthy controls

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ARTICLE INFO

*Article history:*
Received 25 October 2024
Revised 8 January 2025
Accepted 8 January 2025

ABSTRACT

*Objective:* Due to their medical vulnerability, youngsters with congenital heart disease (CHD) may experience more overwhelming emotions than healthy peers. This multi-informant-based study aims to examine differences between these youngsters and their peers in psychosocial functioning, attachment and emotion regulation.
*Study design:* 217 youngsters (8–18 years) with CHD (53.9 % boys, 46.1 % girls) were compared to 232 healthy controls (52.6 % boys, 47.4 % girls) matched for gender, age and education. Participants and parents completed online self-report questionnaires assessing psychosocial functioning (SDQ), attachment (ECR-RC), and maladaptive Emotion Regulation Strategies (ERS; FEEL-KJ).
*Results:* Based on child's self-reports MANOVA's showed no significant differences between the groups in psychosocial functioning. However, based on parent reports, differences were found between the groups in psychosocial functioning for the total scales and overarching subscales. No differences were found between the groups for insecure attachment to either parent. However, youngsters with CHD and their fathers reported more use of self-devaluation compared to controls ($p = .03$). Other maladaptive ERS (giving up, withdrawal, rumination, aggressive actions) showed no differences.
*Conclusions:* Care interventions for children with CHD should address medical, emotional, and social needs, with a focus on multi-informant evaluations to support emotional well-being. Nurses are important partners in detecting psychosocial difficulties and providing family support. Patient- and family-centered care involves patients, parents and caregivers in the care plan, recognizing their key role, especially as youngsters often perceive their psychosocial health differently than their parents. Although differences were noted compared to the control group, the study's cross-sectional design limits conclusions on evolution with time.


## Introduction

Congenital heart diseases (CHD) are the most prevalent congenital defects, encompassing structural abnormalities of the heart, occurring in 0.8 % of births (Moons et al., 2009). Technological advancements and surgical interventions have enhanced the life expectancy of children with CHD in the recent decades (Mandalenakis et al., 2020). However, medical procedures and treatments are stressful (Lerwick, 2016), possibly contributing to impaired emotional and psychosocial functioning (Cassedy et al., 2023; Monti et al., 2018). These hospitalizations and (early) separation from parents, potentially affect the child's attachment (Hsiao et al., 2024; Tesson et al., 2019) and development (Berant et al., 2008). Coping with stress and regulating emotions are crucial in the development from childhood to adulthood (Compas et al., 2017). Emotion regulation (ER) refers to managing and modifying emotional reactions to maintain psychological well-being (Gross, 1998). A tripartite model of familial influence describes pathways through which familial factors exert an impact on children's ER, which consequently impacts outcomes of maladjustment and wellbeing (Morris et al., 2007). According to this theoretical based framework, building secure attachment relations is challenged and disturbances in ER can be expected when children are separated from their parents (Morris et al., 2007). Previous research has mainly focused on problems in neurocognitive functioning of patients with CHD (Sterken et al., 2015), with limited exploration of their psychosocial vulnerability

*Abbreviations:* ANOVA, Analysis of Variance; CHD, Congenital Heart Disease; ECR-RC, Experiences in Close Relationships Scale-Revised Child version; EMA, Ecological Momentary Assessment; ER, Emotion Regulation; ERS, Emotion Regulation Strategies; FEEL-KJ, Fragebogen zur Erhebung der Emotionsregulation bei Kindern und Jugendlichen; MANOVA, Multivariate Analysis of Variance; NICU, Neonatal Intensive Care Unit; SDQ, Strengths and Difficulties Questionnaire.

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https://doi.org/10.1016/j.pedn.2025.01.006
0882-5963/

compared to healthy controls (Abda et al., 2019; J. L. Jackson et al., 2018) and this research is often based on parental reports (Dahlawi et al., 2020). Some studies have indicated that children with CHD experience lower quality of life and impaired psychosocial functioning (Utens et al., 1993) compared to healthy peers.

Also in pediatric hospital settings, parents are often used as an important source of information on their children's physical, behavioral and emotional wellbeing. Healthcare providers should be aware that care for CHD patients also includes care for their emotional health (Savaş et al., 2024). Nurses play a crucial role both in observing and reporting difficulties as they have a close relationship both with patients and parents. They are often the first to be taken into confidence if parents have a concern. Moreover, they are perfectly placed to be the first to provide support and tips. However, there is little insight into the emotional processes of young CHD patients (Wierenga et al., 2017). This multi-informant study aims to examine whether youngsters with a CHD experience more psychosocial difficulties, differ on insecure attachment towards parents, and use more maladaptive ER strategies in comparison to healthy controls.

## Methods

This observational cross-sectional study used multi-informant reports from youngsters with CHD and their parents, compared with healthy controls. The child-centred EQUATOR research guideline (Foster et al., 2024) was implemented and ethical approval was granted by the Medical Ethics Committee of Ghent University Hospital (BC-07779).

### *Participants and procedure*

**Patient group:** Percutaneously and/or surgically treated CHD patients aged 8 to 17 were selected from the pediatric cardiology database of Ghent University Hospital (Belgium). Medical data was collected by the cardiologist based on the Electronic Patient File. Exclusion criteria included associated complex non-cardiac pathology, genetic abnormalities, intellectual disability, or insufficient knowledge of Dutch. Potential CHD families were invited to participate via letter and/or email.

**Control group:** Youngsters with CHD were asked to refer a friend of the same gender, age, and education level. Due to an insufficient number of self-provided controls, further recruitment was conducted via an online call on the websites of Ghent University and Ghent University Hospital.

### *Questionnaires*

Upon agreeing to participate, families received an active informed consent document and a unique login code for a secured website or printed questionnaires. All respondents (youngsters, mothers, and fathers) were given unique codes to enable post hoc matching. Depending on the age of the youngster, parents and/or youngsters completed different validated and standardized questionnaires.

#### *Questionnaires completed by the youngsters and/or the parents*

**SDQ:** The psychosocial functioning of youngsters was evaluated using the Dutch Strengths and Difficulties Questionnaire (SDQ). Parents completed the version for 4- to 17-year-olds (Goedhart et al., 2003), while youngsters aged 11 or older filled in the version for 11- to 17-year-olds (Treffers & van Widenfelt, 2000). Twenty items measure four domains of psychosocial difficulties: hyperactivity/inattention (e.g., “Restless, overactive, cannot stay still for long”), emotional problems (e.g., “Many fears, easily scared”), conduct problems (e.g., “Often has temper tantrums or hot tempers”), and social problems (e.g., “Picked on or bullied by others”). In addition to a total scale (which includes 5 prosocial items), two overarching subscales are differentiated: externalizing (hyperactivity/inattention and conduct scales) and internalizing difficulties (emotional and social scales). Items are rated on a 3-point scale (0 = not true, 1 = somewhat true, 2 = certainly true), with higher scores indicating greater difficulties. The internal consistency is good (Maurice-Stam et al., 2018).

**FEEL-KJ:** Youngsters' use of Emotion Regulation Strategies (ERS) was measured using the Dutch FEEL-KJ (Fragebogen zur Erhebung der Emotionsregulation bei Kindern und Jugendlichen) (Braet et al., 2014). This questionnaire assesses 15 strategies (maladaptive, adaptive, and external) across three emotions: anxiety, sadness, and anger, through 90 items. Youngsters respond on a 5-point Likert scale, ranging from 1 (almost never) to 5 (almost always), with higher scores indicating greater use of an ERS. In this study, only maladaptive ERS were included, comprising five strategies: (1) giving up (e.g., “I don't want to do anything”), (2) aggressive actions (e.g., “I get into a quarrel with others”), (3) withdrawal (e.g., “I don't want to see anyone”), (4) self-devaluation (e.g., “I blame myself”), and (5) rumination (e.g., “I cannot get it out of my head”). A total maladaptive score was also calculated. Psychometric evaluation of the FEEL-KJ revealed a good internal consistency (Cracco et al., 2015).

#### *Questionnaire completed by the youngsters only*

**ECR-RC:** Attachment to parents was assessed using the Dutch self-report questionnaire ‘Experiences in Close Relationships Scale-Revised Child version (ECR-RC)’ (Brenning et al., 2011). Twelve items for each parent are rated on a 7-point Likert scale, ranging from 1 (strongly disagree) to 7 (strongly agree). This study used both subscales: Attachment Anxiety (e.g., ‘I worry that my father/mother does not really love me’) and Attachment Avoidance (e.g., ‘I prefer not to tell my father/mother how I feel deep down’). A higher score indicates a more insecure attachment pattern, while a lower score reflects a more secure attachment. Both subscales show high internal consistency (Sibley et al., 2005).

### *Statistical analysis*

Statistical analysis was performed using SPSS (version 28) (IBM, Armonk, NY). Comparisons between the patient and control groups were made regarding demographics and outcome measures. Nominal data were analyzed using $\chi^2$ statistics. Univariate ANOVAs were conducted with group as the between-subjects factor and total scores on the SDQ and FEEL-KJ as dependent variables. MANOVAs were employed with group as the between-subjects factor and specific subscale scores of the SDQ, ECR-RC, and FEEL-KJ as dependent variables. The significance level was set at $p < .05$.

## Results

### *Sample characteristics*

After identification of 592 eligible youngsters from the congenital cardiology database, recruitment took place between November 2020 and December 2022. In total, 449 families participated. The patient group consisted of 217 participants ($N = 117$ boys, $N = 100$ girls; response rate 36,7 %) with various CHDs, prenatally or postnatally diagnosed (see Table 1). CHD was categorized into three categories, based on disease severity (Warnes et al., 2001).

Pediatric CHD patients aged 8 to 17 were recruited, two patients turned 18 years old during study participation. 185 patients had at least one surgical intervention, with the age at the first intervention ranging from 1 day to 14.3 years; 32 patients only underwent percutaneous treatment. Patients were matched for age, gender and education level with 232 healthy controls ($N = 122$ boys, $N = 110$ girls).

There was no difference between the groups in gender ($p = .778$), age ($p = .827$), and education ($p = .078$). Due to their cardiac disease, more children with CHD were admitted to the Neonatal Intensive Care

**Table 1**
Characteristics of the patient group ($N = 217$).

| Characteristics | N | (%) |
|---|---|---|
| Heart defect* | | |
| **Simple complexity** | **73** | **(33.6)** |
| Atrial Septal Defects (ASD I, ASD II or sinus venosus ASD) | 39 | (18.0) |
| Ventricular Septal Defect (VSD) | 26 | (12.0) |
| Aortic valve stenosis | 8 | (3.7) |
| **Moderate complexity** | **87** | **(40.1)** |
| Atrioventricular Septal Defect (AVSD) | 8 | (3.7) |
| Coarctation of the Aorta (CoA and CoA with VSD) | 33 | (15.2) |
| d-Transposition of the Great Arteries (d-TGA) | 20 | (9.2) |
| Pulmonary valve stenosis | 10 | (4.6) |
| Coronary abnormality | 3 | (1.4) |
| Ebstein anomaly | 4 | (1.8) |
| Total Anomalous Pulmonary Venous Return (TAPVU) | 2 | (0.9) |
| Other cardiac abnormalities | 7 | (3.2) |
| **Severe complexity** | **57** | **(26.3)** |
| Mitral valve stenosis | 5 | (2.3) |
| Double-outlet Right Ventricle (DORV) | 3 | (1.4) |
| Truncus arteriosus | 2 | (0.9) |
| Pulmonary atresia with intact septum | 5 | (2.3) |
| Tetralogy of Fallot (TOF and TOF with MAPCAs) | 21 | (9.7) |
| Hypoplastic Left Heart Syndrome (HLHS) | 15 | (6.9) |
| Aortic valve stenosis | 5 | (2.3) |
| Other cardiac abnormalities | 1 | (0.5) |
| Timing diagnosis | | |
| Prenatal | 28 | (13,3) |
| Postnatal | 188 | (87.1) |
| Missing | 1 | (0.5) |

*Note.* $N$ = number; *heart defect severity: in the absence of a comprehensive assessment of disease severity in children with CHD, the child cardiologists generated a classification system, similar to the one used in adults with CHD (Task Force 1 of the 32nd Bethesda conference of the American College of Cardiology, 2001). The Bethesda disease complexity classification categorizes patients into 3 groups: simple, moderate, and severely complex congenital heart defects, based solely on the anatomic complexity.

Unit (NICU) ($p < .001$). Demographic data for the youngsters and their parents can be found in Tables 2 and 3.

*Reliability of the questionnaires*

**SDQ.** In this study Cronbach's α for the total scale was 0.77 (child-report), 0.85 (mother-report), and 0.84 (father-report). For the internalizing scale, Cronbach's α was 0.69 (child-report), 0.78 (mother-report), and 0.77 (father-report), and for the externalizing scale, it was 0.77 (child-report) and 0.80 (mother- and father-report respectively). Cronbach's α for the four difficulties subscales ranged between 0.40 and 0.79 (child-report), 0.54 and 0.81 (mother-report), and 0.58 and 0.80 (father-report). Due to acceptable but lower internal consistency, careful interpretation is required. In total, 239 youngsters, 399 mothers and 301 fathers completed the SDQ.

**FEEL-KJ.** Cronbach's α for the total maladaptive scale was 0.84 (child-report), 0.85 (mother-report), and 0.84 (father-report). The five subscale coefficients ranged from 0.66 to 0.83 (child-report), 0.68 to 0.86 (mother-report), and 0.66 to 0.88 (father-report). A total of 316 youngsters, 340 mothers, and 253 fathers completed the FEEL-KJ.

**ECR-RC.** Cronbach's α for Attachment Anxiety towards mother was 0.88 and 0.94 for Attachment Anxiety towards father, and for Attachment Avoidance, it was 0.87 for both Attachment Avoidance towards mother and towards father. A total of 354 youngsters completed the ECR-RC.

*Analysis of variance*

The (M)ANOVAs comparing the patient and the control group on psychosocial functioning, attachment, and ERS are presented in Table 4.

*Psychosocial functioning*

Self-reports by youngsters on total psychosocial functioning did not show a significant difference between the patient and control group ($p = .636$). Upon closer examination, youngsters with CHD did not differ from the control group regarding internalizing or externalizing problems, nor on any of the four individual subscales (emotional problems, hyperactivity, conduct problems and social problems).

However, in contrast, parents reported a significantly higher presence of psychosocial difficulties in their children with CHD. Reports from both mothers and fathers revealed a significant difference between the groups in total psychosocial functioning ($p < .001$ for mothers and $p = .007$ for fathers). Looking more closely at the differences on internalizing and externalizing problems reported by both parents, a difference between the groups was found ($p < .001$ for mothers and $p = .021$ for fathers): CHD parents reported more internalizing

**Table 2**
Demographic data of the youngsters in the two study groups.

| Characteristics | | Patient group N = 217 | | Control group N = 232 | | p |
|---|---|---|---|---|---|---|
| | | Mean | ±SD | Mean | ±SD | |
| Age (years) | | 12.20 | ±2.86 | 12.26 | ±2.96 | 0.827 |
| | | N | (%) | N | (%) | |
| | 8–10 years | 71 | (32.7) | 76 | (32.8) | |
| | 11–12 years | 50 | (23.1) | 49 | (21.1) | |
| | 13–15 years | 61 | (28.1) | 75 | (32.3) | |
| | ≥ 16 years | 35 | (16.1) | 32 | (13.8) | |
| | | N | (%) | N | (%) | |
| Gender | | | | | | 0.778 |
| | Male | 117 | (53.9) | 122 | (52.6) | |
| | Female | 100 | (46.1) | 110 | (47.4) | |
| Education | | | | | | 0.078 |
| | Primary School | 93 | (42.9) | 103 | (44.6) | |
| | Special Needs Education | 11 | (5.1) | 4 | (1.7) | |
| | Secondary school | | | | | |
| | ASO – academic track | 72 | (33.2) | 98 | (42.4) | |
| | TSO – technical track | 23 | (10.6) | 15 | (6.5) | |
| | KSO – arts track | 1 | (0.5) | 1 | (0.4) | |
| | BSO – craft track | 11 | (5.1) | 5 | (2.2) | |
| | Other – (e.g. going to school in other country) | 6 | (2.8) | 5 | (2.2) | |
| Admission to NICU | Yes | 113 | (52.6) | 26 | (11.4) | <0.001*** |

*Note.* $^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$. SD = Standard Deviation; $N$ = number; NICU = Neonatal Intensive Care Unit.

**Table 3**
Demographic data of the parents of the patient and the control group.

| Characteristics | Patient group Father $N = 147$ N (%) | Patient group Mother $N = 189$ N (%) | Control group Father $N = 155$ N (%) | Control group Mother $N = 210$ N (%) |
|---|---|---|---|---|
| Highest education level obtained | | | | |
| Primary School (incomplete) | 0 (0.0) | 0 (0.0) | 1 (0.6) | 0 (0.0) |
| Primary School (completed) | 0 (0.0) | 1 (0.5) | 0 (0.0) | 1 (0.5) |
| Secondary school (incomplete) | 3 (2.0) | 8 (4.2) | 1 (0.6) | 1 (0.5) |
| Secondary school (completed) | 49 (33.3) | 42 (22.2) | 32 (20.6) | 19 (9.0) |
| College (short type) | 38 (25.9) | 63 (33.3) | 40 (25.8) | 60 (28.6) |
| College (long type) | 22 (15.0) | 36 (19.0) | 13 (8.4) | 27 (12.9) |
| University | 33 (22.4) | 38 (20.1) | 66 (42.6) | 99 (47.1) |
| Information missing | 0 (0.0) | 0 (0.0) | 2 (1.3) | 1 (0.5) |
| Another type of highest education | 2 (1.4) | 1 (0.5) | (0.0) | 2 (1.0) |
| Occupation | | | | |
| Worker | 22 (15.0) | 20 (10.6) | 12 (7.7) | 12 (5.7) |
| Employe | 68 (46.3) | 92 (48.7) | 77 (49.7) | 114 (54.3) |
| Self-employed: | | | | |
| Own business | 29 (19.7) | 17 (9.0) | 25 (16.1) | 20 (9.5) |
| In another ones business | 3 (2.0) | 4 (2.1) | 4 (2.6) | 3 (1.4) |
| Government employe | 22 (15.0) | 41 (21.7) | 32 (20.6) | 57 (27.1) |
| Housework | 1 (0.7) | 4 (2.1) | 0 (0.0) | 2 (1.0) |
| Student | 0 (0.0) | 2 (1.1) | 0 (0.0) | 0 (0.0) |
| Disabled | 0 (0.0) | 6 (3.2) | 1 (0.6) | 0 (0.0) |
| Retired | 1 (0.7) | 1 (0.5) | 2 (1.3) | 1 (0.5) |
| Unemployed | 0 (0.0) | 2 (1.1) | 0 (0.0) | 1 (0.5) |
| Another kind of employment | 1 (0.7) | 0 (0.0) | 2 (1.3) | 0 (0.0) |

Note. $N$ = number.

($p < .001$ for mothers and $p = .015$ for fathers) and externalizing problems in their children ($p < .001$ for mothers, and $p = .026$ for fathers). Additionally, differences between the two groups were detected regarding the four difficulties scales, but only for the mothers' reports ($p < .001$) and not for the fathers' reports ($p = .102$).

Based on father reports, further analysis demonstrated significantly higher scores for emotional problems ($p = .019$) and hyperactivity ($p = .042$) in children with CHD compared to their healthy peers. Mother reports confirmed the findings for emotional problems ($p < .001$) and hyperactivity ($p < .001$) and also revealed a higher presence of social problems ($p < .001$). According to both fathers and mothers, no significant differences between both the two groups were found for conduct problems.

*Attachment*

There was no difference between the two groups in youngster's self-reports on insecure attachment to their mothers ($p = .944$) or fathers ($p = .138$).

*Maladaptive emotion regulation*

Using youngsters' self-reports, there was no difference between the patient and control groups regarding the total use of maladaptive ERS ($p = .181$). However, a trend-significant difference was found for the maladaptive ERS subscales ($p = .051$). A more detailed analysis of the subscales revealed that patients tended to report a higher use of self-devaluation compared to healthy controls ($p = .030$). For the other ERS, giving up, aggressive actions, withdrawal and rumination, no differences were found.

When using parents as informants regarding their youngsters' ERS, the total use of maladaptive ERS did not differ between the patient and the control group, whether reported by mothers ($p = .469$) or fathers ($p = .906$). For mother-reported data, no significant differences

**Table 4**
(M)ANOVAs comparing the patient group and the control group.

| Outcomes | Patient group Mean | Patient group ±SD | Control group Mean | Control group ±SD | *F* | *p* |
|---|---|---|---|---|---|---|
| **YOUNGSTER-REPORTED** | | | | | | |
| ***Psychosocial functioning*** | | | | | | |
| Two overarching subscales: | | | | | **0.093** | **0.911** |
| Internalizing problems | 5.49 | 3.42 | 5.31 | 3.14 | 0.178 | 0.673 |
| Externalizing problems | 6.30 | 3.63 | 6.21 | 3.58 | 0.033 | 0.855 |
| Four domains of difficulties: | | | | | **0.225** | **0.924** |
| Emotional problems | 4.02 | 2.62 | 3.89 | 2.54 | 0.158 | 0.692 |
| Conduct problems | 1.51 | 1.31 | 1.56 | 1.49 | 0.073 | 0.787 |
| Hyperactivity | 4.79 | 2.77 | 4.66 | 2.59 | 0.153 | 0.696 |
| Social problems | 1.54 | 1.57 | 1.43 | 1.24 | 0.382 | 0.537 |
| Total difficulties (ANOVA) | 11.86 | 5.42 | 11.53 | 5.34 | 0.224 | 0.636 |
| ***Attachment towards Mother*** | | | | | **0.058** | **0.994** |
| Anxiety | 9.61 | 5.12 | 9.41 | 5.80 | 0.111 | 0.740 |
| Avoidance | 18.21 | 8.33 | 18.04 | 8.07 | 0.036 | 0.850 |
| ***Attachment towards Father*** | | | | | **1.994** | **0.138** |
| Anxiety | 9.28 | 5.37 | 9.30 | 5.92 | 0.001 | 0.974 |
| Avoidance | 21.08 | 8.13 | 19.42 | 8.06 | 3.407 | 0.066 |
| ***Emotion Regulation*** | | | | | **2.233** | **0.051** |
| ***Strategies*** | | | | | | |
| Giving Up | 14.27 | 4.42 | 14.37 | 4.62 | 0.035 | 0.851 |
| Aggressive Actions | 15.84 | 4.98 | 15.17 | 4.93 | 1.382 | 0.241 |
| Withdrawal | 15.54 | 4.95 | 15.34 | 4.95 | 0.125 | 0.724 |
| Self-Devaluation | 16.64 | 5.06 | 15.36 | 5.21 | 4.753 | 0.030* |
| Rumination | 17.88 | 4.55 | 17.94 | 4.38 | 0.015 | 0.903 |
| ER maladaptive total (ANOVA) | 80.98 | 14.18 | 78.75 | 14.73 | 1.795 | 0.181 |
| **MOTHER-REPORTED** | | | | | | |
| ***Psychosocial functioning*** | | | | | | |
| Two overarching subscales: | | | | | **12.565** | **<0.001*****|
| Internalizing problems | 5.19 | 3.83 | 3.54 | 3.00 | 23.187 | <0.001*** |
| Externalizing problems | 5.19 | 4.00 | 3.96 | 3.40 | 11.073 | <0.001*** |
| Four domains of difficulties: | | | | | **7.883** | **<0.001***** |
| Emotional problems | 3.53 | 2.54 | 2.56 | 2.28 | 16.222 | <0.001*** |
| Conduct problems | 1.32 | 1.53 | 1.12 | 1.38 | 1.900 | 0.169 |
| Hyperactivity | 3.87 | 3.01 | 2.84 | 2.47 | 14.071 | <0.001*** |
| Social problems | 1.66 | 1.85 | 0.98 | 1.24 | 18.813 | <0.001*** |
| Total difficulties (ANOVA) | 10.38 | 6.78 | 7.50 | 5.20 | 22.929 | <0.001*** |
| ***Emotion Regulation*** | | | | | **0.585** | **0.712** |
| ***Strategies*** | | | | | | |
| Giving Up | 14.56 | 4.82 | 13.99 | 4.17 | 1.256 | 0.263 |
| Aggressive Actions | 17.60 | 4.21 | 18.10 | 4.49 | 1.011 | 0.315 |
| Withdrawal | 15.27 | 5.38 | 14.96 | 4.66 | 0.309 | 0.579 |
| Self-Devaluation | 15.52 | 4.17 | 14.97 | 4.03 | 1.423 | 0.234 |
| Rumination | 17.34 | 3.80 | 16.98 | 3.73 | 0.708 | 0.401 |
| ER maladaptive total (ANOVA) | 80.26 | 13.45 | 79.22 | 12.92 | 0.526 | 0.469 |
| **FATHER-REPORTED** | | | | | | |
| ***Psychosocial functioning*** | | | | | | |
| Two overarching subscales: | | | | | **3.927** | **0.021*** |
| Internalizing problems | 4.47 | 3.71 | 3.51 | 3.07 | 5.978 | 0.015* |
| Externalizing problems | 5.28 | 3.90 | 4.30 | 3.70 | 5.021 | 0.026* |
| Four domains of difficulties: | | | | | **1.953** | **0.102** |
| Emotional problems | 3.01 | 2.53 | 2.36 | 2.31 | 5.528 | 0.019* |
| Conduct problems | 1.47 | 1.64 | 1.17 | 1.51 | 2.877 | 0.091 |
| Hyperactivity | 3.78 | 2.82 | 3.13 | 2.71 | 4.155 | 0.042* |
| Social problems | 1.42 | 1.71 | 1.15 | 1.41 | 2.134 | 0.145 |
| Total difficulties (ANOVA) | 9.69 | 6.57 | 7.81 | 5.44 | 7.323 | 0.007** |
| ***Emotion Regulation*** | | | | | **3.448** | **0.005**** |
| ***Strategies*** | | | | | | |
| Giving Up | 13.97 | 4.72 | 13.71 | 4.31 | 0.185 | 0.668 |
| Aggressive Actions | 17.69 | 3.79 | 18.08 | 4.26 | 0.536 | 0.465 |
| Withdrawal | 14.24 | 5.09 | 14.96 | 4.67 | 1.271 | 0.261 |
| Self-Devaluation | 15.49 | 4.11 | 14.39 | 3.55 | 4.827 | 0.029* |
| Rumination | 16.24 | 3.50 | 16.89 | 3.37 | 2.044 | 0.154 |
| ER maladaptive total (ANOVA) | 77.47 | 13.74 | 77.67 | 12.03 | 0.014 | 0.906 |

*Note.* $^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$. SD = Standard Deviation; ER = Emotion Regulation.

were found for any of the five specific ERS. By contrast, father-reported data showed significant differences on the five specific ERS ($p = .005$). Further analysis of the univariate differences demonstrated that fathers reported a significantly higher use of self-devaluation ($p = .029$) among CHD youngsters compared to fathers of youngsters in the control group.

## Discussion

To date, most research on psychological morbidities in youngsters with CHD has focused on neurocognitive, psychiatric and behavioral problems (Kovacs et al., 2022; Mebius et al., 2017). Only recently has the focus shifted to give more attention to their social and emotional functioning (Clancy et al., 2020). Previous research primarily relied on parents as the main source of information, with only a few studies utilizing child self-reports (Dahlawi et al., 2020). Furthermore, many of these studies did not include a healthy control group and merely compared the ratings of the youngsters with normative data (Finkel et al., 2023). We investigated a large sample of 8- to 18-year-olds with CHD and compared them to a matched healthy control group using standardized self-reports as well as parental reports on youngsters' psychosocial functioning.

### *Psychosocial functioning*

First, a notable discrepancy was observed between self-reports and parent-reports regarding psychosocial outcomes. While patients' self-reports showed no differences between CHD and healthy youngsters, both mothers and fathers of CHD patients reported a significantly higher prevalence of psychosocial difficulties, including internalizing and externalizing problems. Both parents noted increased emotional problems and hyperactivity, with mothers additionally reporting more social issues.

These divergent findings on the psychosocial functioning of youngsters with congenital heart disease (CHD) align with previous research. A review of CHD adolescents and adults using self-reports found no differences in emotional functioning compared to healthy controls (A. C. Jackson et al., 2015). However, other studies reported significant differences in social problems and hyperactivity among adolescents with CHD (Schaefer et al., 2013). In contrast, studies using parent-reports found significantly higher scores for internalizing, externalizing, and total psychosocial problems in CHD children and adolescents compared to healthy controls (Abda et al., 2019; Dahlawi et al., 2020). A meta-analysis by Finkel and collegues (Finkel et al., 2023) also reported increased internalizing and total psychosocial problems but not more externalizing symptoms based on parent and self-reports. Thus, while CHD youngsters in this study may not consistently report more psychosocial issues themselves, parent reports suggest potential difficulties that warrant attention, considering different perspectives on psychosocial functioning.

When only one perspective is assessed, potential biases must be considered. Children with CHD may struggle to recognize and articulate their problems and may not communicate openly with their parents or healthcare providers. Conversely, parents may under- or overestimate issues based on their perceptions and stress levels. Parental awareness, knowledge, and the context in which they assess their child's psychosocial functioning also influence their reports. Furthermore, children report from their direct experiential perspective, which can be subjective, while parents interpret behaviors from an external viewpoint. Social desirability may also be a factor (De Los Reyes & Kazdin, 2005).

In addition to examining differences in psychosocial functioning, this study utilized the theoretical family-based framework of Morris et al. (2007) to investigate whether CHD youngsters report more insecure attachment and maladaptive ERS compared to healthy controls.

### *Attachment*

To date, the dynamics of the parent-child relationship in CHD are not well understood (Tesson et al., 2022). Most research relies on observational studies or parent reports in infancy, which limits the interpretation of results due to methodological issues. Most studies found no evidence of widespread disruptions in early caregiving relationships after a CHD diagnosis (Tesson et al., 2022). Similarly, the present study found no differences in children's self-reported attachment to both mothers and fathers compared to healthy controls. The parent-child relationship is unique and requires flexible approaches, especially in a medical context (Jordan et al., 2014). Apart from health, attachment is influenced by various factors, including the child's temperament, parental coping strategies, family resilience, and social support (Bosmans et al., 2020; Morris et al., 2007). Parents of children with CHD may be particularly attuned to their children's psychosocial challenges (Hsiao et al., 2024; Lumsden et al., 2019), striving to maintain a strong emotional bond and offering similar responsiveness, warmth, and consistency as parents of healthy children. These adaptive strategies may lead to attachment quality comparable to that of children without health issues, although further research is needed to confirm this (Bosmans et al., 2020). Although no differences in attachment relationships were found between the patient and control groups in the present study, it remains essential for nurses to remain vigilant for individual cases where attachment processes appear to be challenging.

### *Maladaptive emotion regulation*

According to Morris et al. (2007), differences in psychosocial problems may stem from youngsters' use of maladaptive ERS. Our findings on CHD youngsters' use of maladaptive ERS are mixed. Both self-reports and paternal reports (but not maternal reports) suggest that CHD populations are more likely to engage in self-devaluation compared to healthy peers. Self-devaluation, the tendency to blame oneself for one's experiences, significantly correlates with symptoms of depression and anxiety in adolescents and adults (Garnefski & Kraaij, 2006). The increased use of self-devaluation among CHD youngsters may have important implications for their socio-emotional development.

Contrary to our expectations, none of the other ERS showed significant differences between the patient and control group. Although maladaptive ER strategies in CHD youngsters have not been explicitly studied before, one study (Cassidy et al., 2015) found some evidence of difficulties in emotional functioning (in the context of executive functioning) as reported by children and parents. Additionally, young adult CHD patients have shown difficulties coping with stressors (Mathena et al., 2024). In contrast, studies of CHD toddlers and preschoolers found no significant differences in emotional lability or ERS compared to healthy peers (Stasio et al., 2019; Taylor et al., 2024). However, Stasio et al. (2019) did report lower use of adaptive ER in CHD toddlers based on maternal reports, but not paternal reports. This discrepancy emphasizes the need to include both youngsters and parents in research, as their separate reports may provide complementary insights.

Reflecting on these results, children with CHD and healthy children may develop similar maladaptive ERS due to shared developmental processes and stressors, such as school and social interactions. Both groups are influenced by similar parenting practices, which significantly impact ER through responsiveness and warmth (Lumsden et al., 2019). Additionally, non-health-related stressors like family dynamics, academic pressure, and peer relationships may lead to comparable emotional outcomes. A possible explanation for this finding is that families of patients receive psychosocial support from a multidisciplinary team of healthcare providers within the medical context, helping mitigate maladaptive strategies and balancing emotional outcomes with those of healthy children (Rodrigues et al., 2023). Furthermore, resilience and adaptive coping mechanisms developed by children with CHD may result in similar ERS across both groups. The variability within the CHD population indicates that not all children experience significant emotional and psychological impacts, leading to outcomes statistically similar to those of healthy children. More research on adaptive ERS in larger and more diverse samples would be valuable. Nevertheless, in each patient interaction, nurses play a crucial role in sustaining or facilitating the strengths of youngsters with CHD.

### *Strengths and limitations*

The strengths of this study include the large number of participating families, encompassing both patients and their mothers and fathers. Patients with a range of mild to complex CHD pathologies were recruited, medically representing the average young CHD population. A multi-informant approach facilitated investigation from both parental and child perspectives on psychosocial functioning, attachment, and ER. Additionally, the inclusion of a control group matched for age, sex, and education enables comparisons with healthy peers.

Several limitations should be considered. Research based on voluntary participation is susceptible to selection bias, as vulnerable families may have been reluctant to participate, which could partially explain the results. For feasibility, only a limited number of ERS were examined, restricting the findings to narrow domains. Additionally, both patient and control group were assessed during the COVID-19 pandemic, which may have influenced the results. Due to acceptable but lower internal consistency, cautious interpretation of the SDQ is necessary. Data from children and adolescents were analyzed together; however, adolescence involves significant changes and instability, potentially obscuring age-specific differences. Lastly, the study's cross-sectional design limits conclusions about the temporal relationships between variables.

### *Clinical implications*

As proposed in the scientific statement from the American Heart Association to implement a multidisciplinary care model that ensures the best possible quality of life for patients (Kovacs et al., 2022), this study highlights the importance of identifying psychosocial and emotional problems in young patients with CHD and the need for multiple informants. Nurses and other members of the multidisciplinary team, should not only routinely consider emotional screening and behavioral observation (Utens et al., 2018) but also focus on supporting (daily) stressors, rather than solely addressing health-related issues, to enhance resilience (Cassidy et al., 2021). Tailored care interventions for children with CHD should address medical, emotional, and social needs, with a focus on multi-informant evaluations to support emotional well-being. Patient- and family-centered care involves patients, parents and caregivers in the care plan, recognizing their key role, especially as youngsters often perceive their psychosocial health differently than their parents. Nurses, who maintain close contact with both patients and parents, could assist children and families in navigating the challenges of illness by promoting coping strategies and supporting their emotional well-being. The positive effects of emotion regulation training programs on psychopathology in non-CHD populations (Moltrecht et al., 2021; Wante et al., 2018) suggest that early interventions targeting self-devaluation could similarly benefit CHD families. Therefore, nurses and doctors should have the opportunity to consult with a psychologist to ensure appropriate referrals for psychotherapy, not only to improve emotional outcomes in young patients but also reduce family stress. Nurses could also advocate policies that promote access to specialized mental health resources for these children with CHD in- or outside the hospital doors.

### *Suggestions for future research*

Future research should replicate these findings using a longitudinal design to better understand these dynamics over time, including the role of adaptive emotion regulation strategies (ERS). Healthcare providers should focus on developing and validating outcome measures specific to pediatric CHD populations, including not only long-term physical and developmental outcomes, but also emotional outcomes.

Investigating how a child's health condition affects family functioning, including parental stress and overall family resilience would be beneficial. More concrete, examining the emotion regulation of parents in relation to family climate would be valuable (Morris et al., 2007). Finally, ecological momentary assessment (EMA), collecting real-time data on emotions in a person's natural environment, could be used to further explore the role of regulation of daily stressors in CHD families.

## Conclusion

This study indicates that young patients with CHD perceive their psychosocial health differently and more positively than their parents. Parents of children with CHD reported higher levels of hyperactivity, emotional, and psychosocial problems. Therefore, greater attention should be given to multi-informant evaluations of psychosocial issues to enhance the emotional well-being of children with CHD. Nurses are important partners not only in detecting psychosocial difficulties but also in providing family support.

## Acknowledgements, funding and disclosure statement

The research project reported in this published work was conducted by the author as part of her PhD at the University of Ghent, and employment with the Children's Hospital of the Ghent University Hospital. Funding for this project was thankfully received through VZW Kinderhartfonds. The author(s) have full control of all primary data and agree to allow the journal of publication to review the data if requested. We also declare that we have no conflicts of interest.

## CRediT authorship contribution statement

**Saskia Mels:** Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Fé De Mulder:** Writing – original draft, Visualization, Project administration, Data curation. **Lien Goossens:** Writing – review & editing, Visualization, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Kristof Vandekerckhove:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Data curation, Conceptualization. **Katya De Groote:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Data curation, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Datasets generated and/or analyzed for this study will be available on reasonable request from the corresponding author.

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